Estimating the effectiveness of syndromic screening at airports for Bundibugyo ebolavirus disease

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Abstract

We used a stochastic simulation model to estimate the effectiveness of combined exit and entry airport screening for Bundibugyo ebolavirus disease (BVD), using natural-history parameters from a Bayesian re-analysis of the 2012 Isiro outbreak. For a 12-hour international flight from DRC or Uganda at 86% screening sensitivity, we estimate 65% of infected travellers would arrive undetected (95% CrI: 38–76%). The main driver of this outcome is the relative duration of the the incubation period (approximately 7.7 days) and the onset-to-severe-disease interval (approximately 4 days): most infected travellers board before symptom onset and are undetectable by any syndromic screen, whilst those who are symptomatic progress rapidly to illness severe enough to preclude travel. This is compounded during active epidemic growth, when recently exposed (and therefore pre-symptomatic) cases are overrepresented among travellers. Syndromic airport screening offers limited protection against BVD spread via air travel, and should be complemented by outbreak control at source and strengthened clinical surveillance in receiving countries with high travel connectivity to affected areas.

> **Note**
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> **Data note:** Natural-history parameters (incubation period, onset-to-severe-disease delay) are derived from the 2007 Uganda and 2012 DRC BVD outbreaks. Patient-level data from the current 2026 outbreak are not yet available. The epidemic doubling time range used in the growth-phase sensitivity analysis is estimated from the current outbreak (epiforecasts, 2026). The outbreak was declared a PHEIC on 17 May 2026 (World Health Organization 2026).

1 Introduction

Bundibugyo ebolavirus (BDBV) is one of six known ebolaviruses, previously identified in outbreaks in Uganda (2007) (Towner et al. 2008) and the Democratic Republic of the Congo (DRC; 2012) (Rosello et al. 2015). As of June 2026, an outbreak of Bundibugyo ebolavirus disease (BVD) affecting Ituri Province in DRC and with confirmed spread to Kampala, Uganda has been declared a Public Health Emergency of International Concern (PHEIC) by WHO (World Health Organization 2026). Unlike previous Zaire ebolavirus outbreaks, no approved BVD-specific therapeutics or vaccines exist, and the outbreak is occurring against a backdrop of insecurity, high population mobility, and a large informal healthcare network. Together these features heighten concern about international spread via air travel.

Syndromic airport screening (thermal scanning, symptom questionnaires, and direct clinical assessment) was deployed extensively during the 2014–2016 West Africa Ebola epidemic and again during the 2018–2020 DRC outbreaks of Zaire ebolavirus (Mabey, Flasche, and Edmunds 2014). Its effectiveness, however, is constrained by the natural history of infection: travellers who have been exposed but have not yet developed symptoms will not be febrile at the time of screening, and therefore will not be detected. Whether a traveller is symptomatic at departure or arrival depends on when they were exposed relative to their travel date, on the distribution of the incubation period (infection to symptom onset), and on the time from symptom onset to progression to severe, detectably ill disease.

For BVD specifically, three published incubation period estimates exist and were identified in a recent systematic review (Nash et al. 2024). Wamala et al. (2010) (Wamala et al. 2010) reported a mean of 7.68 days (SD 3.53; 95% CI 7.04–8.32; n = 116) from the 2007 Uganda outbreak. MacNeil et al. (2010) (MacNeil et al. 2010) reported 6.3 days (95% CI 5.2–7.3; n = 24 contacts with confirmed exposure dates) from the same outbreak. Kratz et al. (2015) (Kratz et al. 2015) reported 11.3 days (range 2–27; n = 3) from the 2012 Isiro DRC outbreak.

Total flight durations from DRC or Uganda to international destinations typically range from approximately 10 to 16 hours depending on destination. We model a 12-hour flight as a representative default whilst examining how screening effectiveness varies across a range of flight durations and epidemiological scenarios.

2 Methods

2.1 Simulation model

We used a stochastic individual-based simulation model to evaluate the probability that an infected traveller would be detected at (i) exit screening before departure, (ii) as severely ill during the flight, (iii) at entry screening on arrival, or (iv) remain undetected throughout.

For each simulated traveller, an incubation period t_{inc} and an onset-to-severe-disease interval t_{inf} were drawn independently from Gamma distributions. The time at which a traveller commenced their flight relative to their infection was drawn uniformly from $[0,\ t_{\text{inc}} + t_{\text{inf}} + t_{\text{rec}}]$, where t_{rec} is an individual-specific draw from the onset-to-death Gamma posterior (Funk & Abbott 2026), used as a proxy for the onset-to-recovery interval for asymptomatic cases. Travellers hospitalised before their scheduled departure were excluded.

Exit screening was assumed to detect any traveller who was symptomatic ($t_{\text{flight departure}} > t_{\text{inc}}$) with probability p_{exit}; travellers who became severely ill during the flight or remained symptomatic on arrival were subject to entry screening with probability p_{entry}. A proportion of all infections was assumed to be asymptomatic and remained undetected throughout. Full details are given in Quilty et al. (2020) (Quilty et al. 2020).

2.2 Parameters

Table 1: Model parameters for the BVD default scenario.

Parameter	Value	Source
Flight duration	12 h (default)	DRC/Uganda via connecting hub (representative)
Exit screening sensitivity	86%	Priest et al. 2011 (thermal image scanners)
Entry screening sensitivity	86%	Priest et al. 2011 (thermal image scanners)
Proportion asymptomatic	5%	EVD literature (conservative)
Incubation period mean	7.68 days (95% CI: 7.048.32)	Wamala et al. 2010 (2007 Uganda BVD; n=116)
Incubation period variance	12.46 daysš (SD = 3.53 days)	Wamala et al. 2010
Between-estimate incubation SD	2.6 days (prior SD)	Sample SD across 3 BVD incubation estimates (Nash 2024 / epireview)
Onset severe disease mean	4.03 days (95% CrI: 3.085.46)	Funk & Abbott 2026 posterior median
Onset severe disease variance	13.72 daysš	Funk & Abbott 2026 posterior median

Model parameters are listed in Table 1. The Isiro 2012 line list, the only publicly available BVD line list, contains no exposure dates and was re-analysed using doubly-censored Bayesian methods by Funk & Abbott (2026) (Funk and Abbott 2026), fitting Gamma distributions to the onset-to-severe-disease and onset-to-death delays. The posterior median onset-to-severe-disease mean was 4.0 days (95% CrI: 3.08–5.46 days) and the posterior median SD was 3.7 days (95% CrI: 2.66–5.58 days), corresponding to a Gamma shape of 1.18 and scale of 3.41.

The three published BVD-specific incubation period estimates are described in the Introduction (Wamala et al. 2010; MacNeil et al. 2010; Kratz et al. 2015; Nash et al. 2024). We used the Wamala et al. (2010) estimate as the primary value owing to the largest sample size, and propagated between-study uncertainty using a Normal prior on the incubation mean (mean 7.68 days, SD 2.6 days, truncated above 0.5 days), scaling the variance to preserve the coefficient of variation. This between-study SD of 2.6 days is the sample standard deviation of three point estimates (7.68, 6.3, and 11.3 days), one of which (Kratz et al. 2015) rests on only three cases. A sample SD from three studies carries considerable uncertainty and should be interpreted as an approximate guide to the between-study spread rather than a reliable prior variance; readers should consult the incubation period sensitivity analysis (Figure 4) across the relevant range.

2.3 Uncertainty propagation

For each of 1000 posterior draws sampled at random from `bdbv_posterior_gamma.csv`, we ran the simulation with 2,000 simulated travellers. On each draw, the onset-to-severe-disease delay was taken from the BDBV posterior (draw-specific Gamma mean and variance from the Funk & Abbott 2026 analysis), and the incubation period mean was drawn from the Normal prior described above, with variance rescaled to preserve the coefficient of variation.

2.4 Sensitivity analyses

Four sensitivity analyses were conducted. First, we examined how the proportion undetected varied across flight durations of 1 to 20 hours, with full posterior uncertainty. Second, we explored the joint effect of the proportion of asymptomatic infection (0 to 60%) and the mean incubation period (1 to 14 days) on the proportion undetected for the default 12-hour flight using posterior-median natural-history parameters. Third, to address the limitation that the baseline model assumes a stable epidemic, we re-ran the posterior analysis across epidemic doubling times of 4.5 to 44 days (spanning the 90% credible interval from the epiforecasts BVD 2026 growth-rate estimates (Funk and Abbott 2026)). In a growing epidemic, recently infected individuals are overrepresented among travellers relative to those infected further in the past, so the uniform exposure-time assumption underestimates the pre-symptomatic fraction. Following Gostic et al. (K. M. Gostic, Kucharski, and Lloyd-Smith 2015; K. Gostic et al. 2020), we replaced the uniform draw on $[0, T_{\max}]$ with a truncated-exponential distribution. Each doubling time T_d was converted to an exponential growth rate $r = \ln 2/T_d$, and the time since infection for each simulated traveller was drawn by inverse-CDF sampling: $\;= \ln(1 \;\; U\,(1 \;\; e^{r\,T_{\max}}))/r$, where $U \;\; \mathrm{Uniform}(0,1)$ and $T_{\max} = 2(t_{\text{inc}} + t_{\text{inf}})$. This weights the exposure distribution towards recent infections, as expected under exponential growth. Fourth, we varied exit and entry screening sensitivity independently from 0 to 100% to quantify the marginal contribution of each screening layer.

2.5 Software

All analyses were performed in R (version 4.1; R Core Team 2024). The simulation model is implemented in the `airportscreening` R package, available at github.com/bquilty25/airport_screening_ebola_bvd.

3 Results

3.1 Detection outcomes at 12 hours

Table 2: Estimated screening outcomes for 100 infected BDBV travellers on a 12-hour connecting flight from DRC or Uganda, with 86% sensitivity at exit and 86% sensitivity at entry. Values are percentages with 95% credible intervals.

Detection outcome	Estimate (95% CrI)
Detected at exit screening	26.8 (18.048.1)
Severely ill during flight	0.8 (0.31.9)
Detected at entry screening	6.9 (4.412.4)

Not detected	65.5 (38.476.1)

For a 12-hour connecting flight from DRC or Uganda with 86% screening sensitivity at both departure and arrival, we estimate that 26.8% (95% CrI: 18.0–48.1%) of infected travellers would be detected at exit screening, 0.8% (95% CrI: 0.3–1.9%) would become severely ill during the flight — arriving at the destination airport requiring emergency medical management and representing a distinct public-health import risk even if removed from the "undetected" count — and 6.9% (95% CrI: 4.4–12.4%) would be detected at entry screening (Table 2). In total, 65.5% (95% CrI: 38.4–76.1%) of infected travellers would arrive undetected (Figure 1).

3.2 Effect of flight duration on detection probability

As flight duration increases, the probability of detecting an infected traveller at entry screening rises owing to progression of disease during travel (Figure 2). For the 12-hour connecting itinerary, combined exit and entry screening is estimated to detect 34.5% (95% CrI: 23.9–61.6%) of infected travellers, leaving 65.5% undetected. Even for a 20-hour flight (the longest typical single-airline connection), the proportion undetected remains 63.1% (95% CrI: 37.6–74.8%).

3.3 Detection outcome breakdown

Exit screening accounts for the largest share of detected cases; severely ill during flight and entry-detected cases together represent a small fraction, reflecting the short window between symptom onset and severe disease relative to the flight duration (Figure 3).

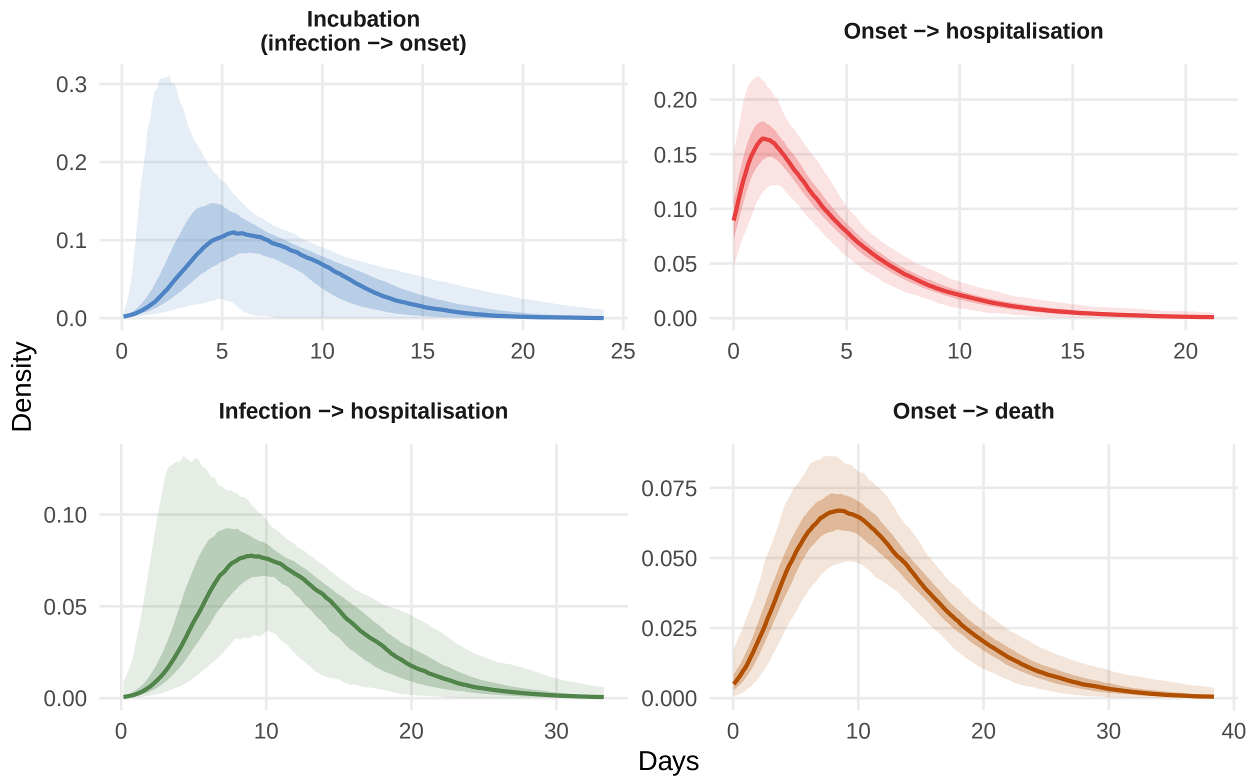

Figure 1: Posterior distributions of natural-history period durations for BVD. Each panel shows densities from 500 joint draws: the onset-to-hospitalisation and onset-to-death delays are drawn from the Bayesian BVD posterior (Funk & Abbott 2026); the incubation period (infection to symptom onset) is drawn from a Normal prior on the mean (7.68 days, SD 2.6 days, truncated > 0.5) with CV preserved (Gamma shape fixed at 4.73). Shaded ribbons show 50% (dark) and 95% (light) pointwise credible bands across the 500 density curves; the wide light ribbon for the incubation period reflects draws with means as low as ~2 days at the 2.5th percentile of the prior. The onset-to-hospitalisation distribution has shape ~1.2 (nearly exponential), reflecting the limited information in the Isiro 2012 line list.

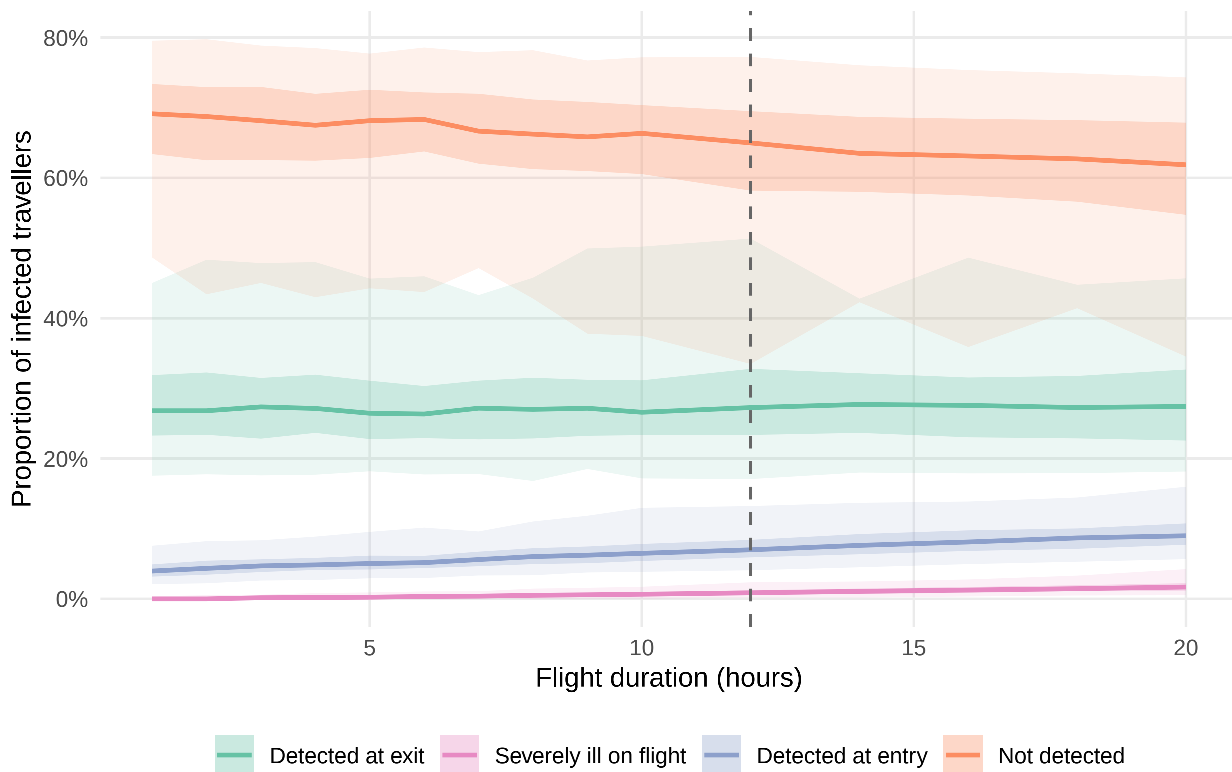

Figure 2: Detection probabilities as a function of flight duration, with uncertainty propagated across 1,000 posterior draws of the BVD onset-to-hospitalisation delay. Ribbons show 50% (dark) and 95% (light) credible intervals. The vertical dashed line marks 12 hours, the approximate DRC/Uganda connecting travel time to international destinations.

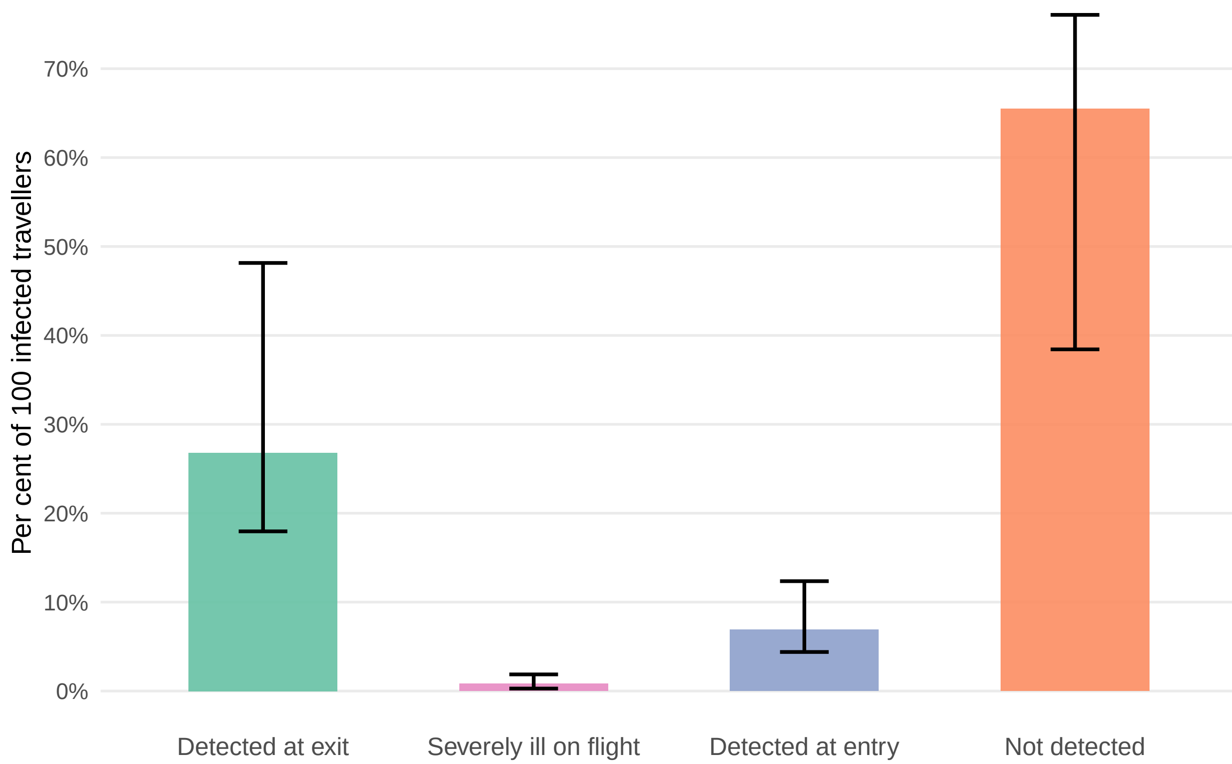

Figure 3: Estimated number of detected and undetected infected travellers per 100 infected travellers departing DRC or Uganda on a 12-hour connecting itinerary, stratified by detection outcome, with 95% credible intervals (whiskers).

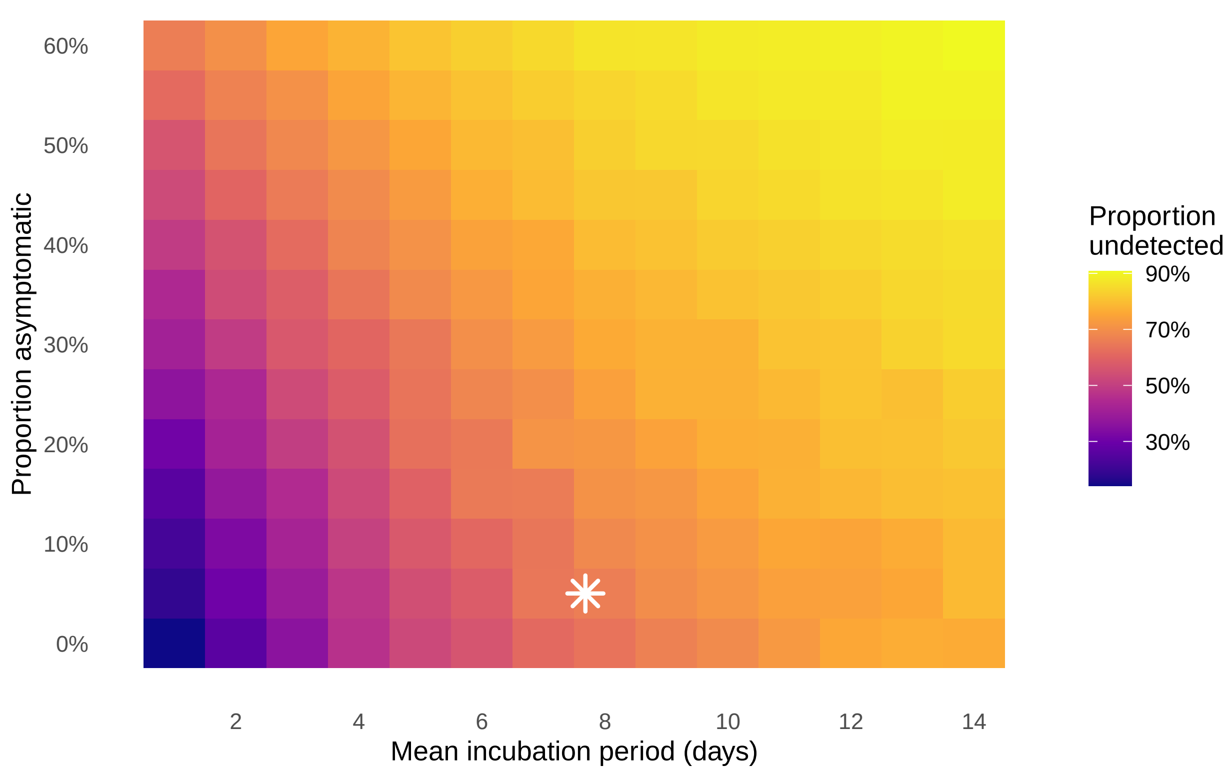

Figure 4: Sensitivity analysis: proportion of infected travellers undetected as a function of the mean incubation period (x-axis) and proportion of asymptomatic infection (y-axis), for a 12-hour flight with posterior-median BVD onset-to-hospitalisation parameters. The star marks the primary BVD scenario (mean incubation 7.68 days, 5% asymptomatic). Full uncertainty propagation across the incubation period is shown in the main analysis.

3.4 Sensitivity analysis: asymptomatic proportion and incubation period

The primary driver of the undetected fraction is the incubation period (Figure 4): even at the conservative BVD default of 5% asymptomatic infection, the proportion undetected ranges from 18–79% across incubation periods of 1–14 days, with longer incubation periods allowing more travellers to depart before symptom onset. Truly asymptomatic EVD infection is rare, but paucisymptomatic presentations that fail to trigger syndromic screening cannot be excluded. If the asymptomatic fraction were as high as 40%, the undetected fraction would rise to 49–86% across the same incubation range, but this is considered an unlikely upper bound for BVD.

3.5 Sensitivity analysis: epidemic growth phase

The proportion of infected travellers going undetected increases as the epidemic doubling time shortens (Figure 5). If the true doubling time falls within the initial epiforecasts 90% credible interval (13.8–22.8 days), the model estimates a corresponding undetected fraction of approximately 69–70%. Under the more pessimistic latest-estimate lower bound (4.5 days), the undetected fraction rises to 80% (95% CrI: 44–91%), compared with 65% under a stable epidemic. This pattern is consistent with the general principle that screening performs less well during periods of rapid epidemic growth, when pre-symptomatic cases are overrepresented among travellers.

3.6 Sensitivity analysis: screening sensitivity at exit and entry

Exit screening sensitivity has a larger marginal effect on the undetected fraction than entry screening sensitivity, because it acts on a larger at-risk window: any traveller already symptomatic at departure can be caught at exit, whereas entry screening only adds value for those who become symptomatic during the flight or who were missed due to imperfect screening at exit (Figure 6). When exit screening is absent

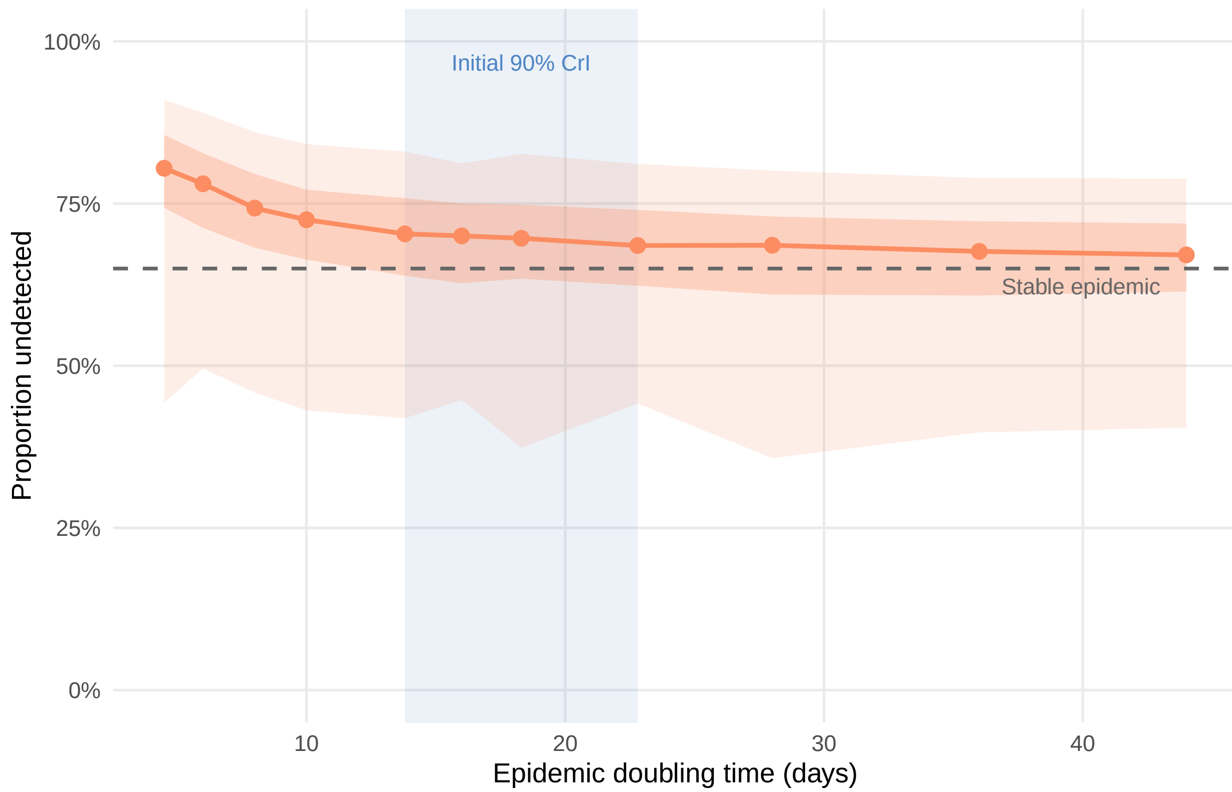

Figure 5: Effect of epidemic doubling time on the proportion of infected travellers going undetected. Each point shows the median across 500 posterior draws; ribbons show 50% (dark) and 95% (light) credible intervals. The shaded band marks the 90% credible interval for the initial doubling time of the current BVD outbreak (13.8–22.8 days; epiforecasts, 2026). The rightmost marker (dashed vertical) represents a stable epidemic (growth rate = 0; uniform exposure distribution).

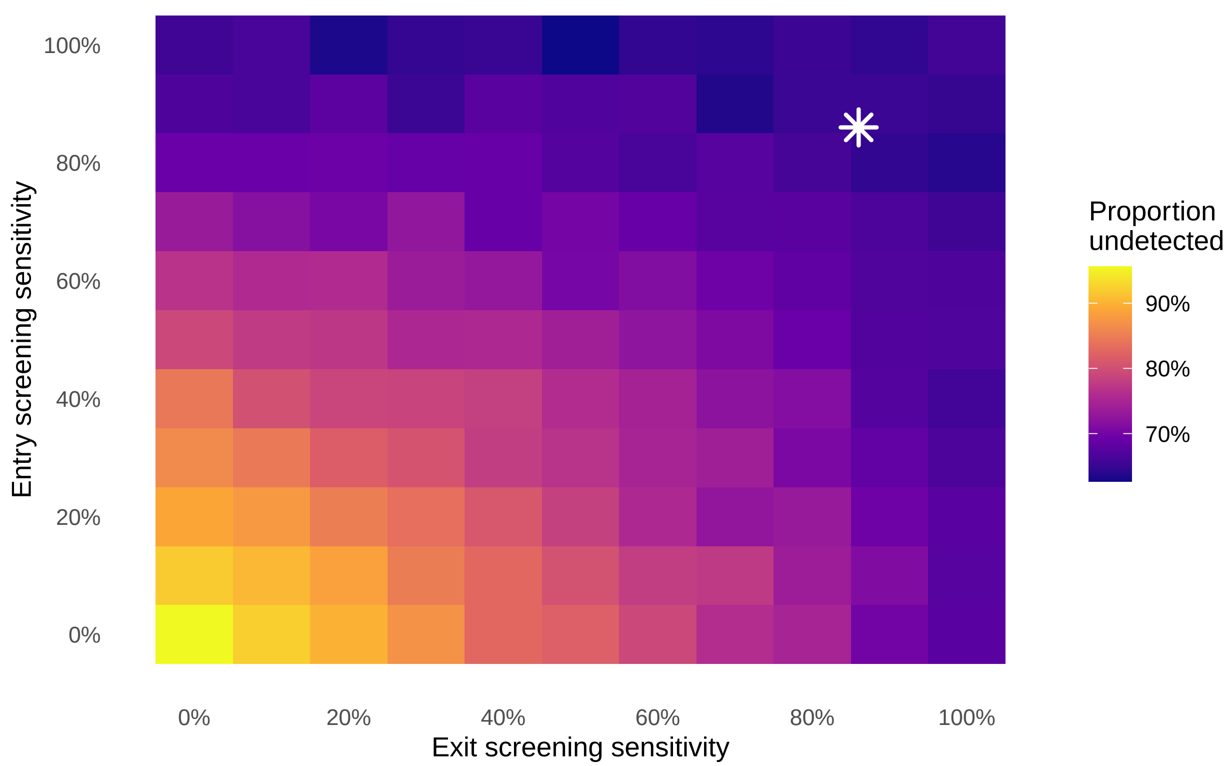

Figure 6: Joint sensitivity analysis: proportion of infected travellers undetected as a function of exit screening sensitivity (x-axis) and entry screening sensitivity (y-axis), for a 12-hour flight with posterior-median BVD natural-history parameters and stable-epidemic exposure distribution. The star marks the primary scenario (86% at both stages). Diagonal dashed lines indicate contours of equal combined coverage.

entirely (sensitivity 0%) but entry screening is maintained at 86%, 67% of travellers remain undetected, compared with 73% when entry screening is absent and exit is maintained at 86%. This asymmetry reflects the longer opportunity for symptom detection that the pre-flight period provides relative to the flight itself.

4 Discussion

At 86% sensitivity on a 12-hour international flight, combined syndromic exit and entry screening will fail to detect approximately 65% of infected travellers (95% CrI: 38–76%). The relatively short onset-to-severe-disease interval (mean approximately four days) means that symptomatic travellers progress to detectable illness quickly, yet the incubation period (approximately 7.7 days Wamala et al. (2010)) means that most infected travellers will board before symptom onset.

The picture is consistent with prior modelling for Zaire ebolavirus and other viral haemorrhagic fevers. Mabey et al. (2014) (Mabey, Flasche, and Edmunds 2014) estimated that exit screening during the 2014 West Africa ZEBOV epidemic would detect only 7–18% of infected travellers on typical long-haul routes, with combined exit and entry screening performing marginally better; the primary driver of failure was departure during the incubation period. Pitman et al. (2005) (Pitman et al. 2005) showed that even under the assumption of 100% screening sensitivity, entry-only screening for SARS would detect approximately one quarter of infected arrivals, with realistic instrument sensitivity (30–50%) reducing this to fewer than one in ten. Gostic et al. (2015) (K. M. Gostic, Kucharski, and Lloyd-Smith 2015) demonstrated that screening effectiveness is primarily determined by the ratio of the incubation period to the onset-to-severe-disease interval and by the proportion of asymptomatic infection; a subsequent application to COVID-19 estimated that even under best-case assumptions, combined departure and arrival screening would detect a median of only 30% (95% interval 10–53%) of infected travellers in a growing epidemic (K. Gostic et al. 2020). Our BVD estimates are of a similar order to those for ZEBOV and are somewhat more favourable than the COVID-19 results under low-asymptomatic assumptions, consistent with the relatively short BDBV incubation period and the low observed asymptomatic fraction. Across the plausible range for BVD, the incubation period is the dominant source of variation in the undetected fraction; the asymptomatic fraction is secondary, given the low rates observed for EVD.

This analysis has several limitations. The BVD-specific evidence base for the incubation period is sparse (only three estimates exist (Wamala et al. 2010; MacNeil et al. 2010; Kratz et al. 2015), as described in the Introduction, and the Kratz et al. (2015) estimate rests on only three cases), and the Isiro 2012 line list comprises only 52 cases. The resulting posterior distributions, whilst appropriately wide, are based on a small sample. The Gamma shape and scale posteriors for the onset-to-admission delay have wide 95% credible intervals, and this uncertainty propagates directly into our screening effectiveness estimates. The model also assumes a single uniform flight duration of 12 hours; in reality, travellers from DRC and Uganda take a variety of itineraries with different durations, layovers, and connection points depending on destination. Our flight-duration sensitivity analysis (Figure 2) demonstrates that outcomes vary only modestly across the 10–16-hour range typical of connecting itineraries, so this simplification does not materially affect the conclusions. The assumed 86% screening sensitivity is derived from a study of thermal image scanning performance during influenza border screening (Priest et al. 2011) and was used in the original 2019-nCoV (later COVID-19) screening model (Quilty et al. 2020); actual sensitivity for EVD symptom detection in field conditions may differ substantially, given differences in fever thresholds, environmental conditions, operator training, and patient cooperation. Because the assumed sensitivity is not BDBV-specific, the screening sensitivity analysis (Figure 6) should be treated as a primary result alongside the central estimates, not as a secondary sensitivity check. Post-arrival healthcare-seeking behaviour is also not captured; infected travellers who develop symptoms after arrival are likely to seek medical care and, if BDBV is considered, to be diagnosed, representing an important additional detection pathway not captured here.

The model removes travellers from the departing pool once they have progressed to severe disease, treating onset-to-severe-disease as equivalent to the threshold for being too ill to travel. In DRC and Uganda, however, hospitalisation can lag clinical deterioration by hours to days owing to health-system access constraints, geographic barriers, and health-seeking norms. The effective "too ill to travel" threshold is therefore likely to be later than the biological severe-disease onset assumed here, meaning the model may underestimate the proportion of symptomatic travellers who successfully board, an anti-conservative bias in this specific direction.

The model does not account for traveller selection effects. During an active outbreak, some infected individuals may travel specifically to seek care that is unavailable locally, biasing the departing caseload towards symptomatic presentations; conversely, awareness of the outbreak, border communication campaigns, or voluntary self-isolation may deter others, reducing the infected fraction among travellers overall. These pressures operate in opposite directions and are difficult to quantify without data on travel volumes and health-seeking behaviour specific to the outbreak context. The baseline assumption, that infected travellers are drawn at random from the infected population with a uniform exposure-age distribution, is a simplification that may not hold in practice.

The simulation model also assumes that exposure times are distributed uniformly across the natural history window, which approximates a stable (non-growing) epidemic. In an actively growing outbreak, as the current DRC BDBV situation likely is, the distribution of infection ages among travelling cases is skewed towards recent exposure, because recent cases outnumber older ones. Under this scenario, a greater fraction of departing travellers will be pre-symptomatic, and the proportion undetected will be higher than our central estimates suggest. Gostic et al. (2015, 2020) (K. M. Gostic, Kucharski, and Lloyd-Smith 2015; K. Gostic et al. 2020) showed that this epidemic-phase effect can reduce detection, and our growth-phase sensitivity analysis (Figure 5) demonstrates a similar pattern for BVD: under the initial 90% credible interval for the current BVD outbreak doubling time (13.8–22.8 days; epiforecasts 2026 (Funk and Abbott 2026)), the undetected fraction increases relative to the stable-epidemic baseline, and under the most pessimistic scenario (doubling time 4.5 days) may be higher still. Our central estimates should therefore be interpreted as a lower bound on the undetected fraction during an active growth phase. The model does not include questionnaire-based exposure risk screening, which forms part of real-world EVD port-of-entry protocols. Since BVD transmission requires direct contact with a symptomatic or deceased case, travellers with known exposure history should, in principle, be detectable by structured risk questionnaires even before fever onset. Incorporating this pathway would increase modelled detection rates above the syndromic-screening-only estimates here. Finally, the model conditions on a traveller already being infected and therefore estimates the sensitivity of screening, defined as the probability of detecting an infected traveller, rather than predictive values. Translating these results into positive or negative predictive values, or into expected counts of imported cases, would require assumptions about the prevalence of infection among travellers departing affected areas, which is highly uncertain during an active outbreak and outside the scope of this analysis.

Syndromic airport screening for BVD is unlikely to detect most infected travellers. This conclusion is consistent with the PHEIC temporary recommendations issued by WHO (World Health Organization 2026), which advise exit screening in affected countries but explicitly state that entry screening outside the affected region is not considered necessary. Resources directed at airport screening in receiving countries carry an opportunity cost: equivalent investment in contact tracing capacity, isolation infrastructure, and healthcare worker protection in DRC and Uganda would more directly reduce onward transmission at the source. For receiving countries, the more actionable layer of protection is clinician preparedness: healthcare providers in low-incidence settings should consider BVD in the differential diagnosis of febrile illness with a travel history from affected regions, supported by clear referral pathways to highconsequence infectious disease units, pre-agreed diagnostic protocols at those centres, and timely communication between national public health authorities and point-of-care clinicians.

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